MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED, STANDBY TYPES I, II, AND III

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the performance, design, and test requirements for Standby Indicated Airspeed Indicators Types I, II, and, III. The airspeed indicators develop indicated airspeed as a function of the difference between pitot and static pressures. The airspeed is to be displayed on an integrally lighted, nominal 2-inch dial covering the airspeed range as applicable to indicator type.

1.2 Classification. The types of airspeed indicators covered by this specification are as follows:

	Type	Range Airspeed	Lighting	Case Shape	Dial Display
a.	I	30-650 Knots	White (IPL)	Square	1 pointer
b.	II	60-850 Knots	White (IPL)	Round	1 pointer
c.	III	20-600 Knots	White (IPL)	Square	2 pointer

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	- Boxes, fiberboard

Military

MIL-P-116	- Preservation-Packaging, Methods of
DOD-D-1000/1	- Drawings Engineering and Associated Data
MIL-B-5087	- Bonding, Electrical, and Lighting Protection for Aerospace Systems
MIL-W-5088	- Wiring, Aircraft, Installation of
MIL-E-5400	- Electronic Equipment, Airborne, General Specification for
MIL-C-5541	- Chemical Films for Aluminum and Aluminum Alloys
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	- Anodic Coatings, for Aluminum Alloys and Aluminum
MIL-C-14806	- Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	- Test Procedures: Preproduction, Acceptance and Life for Aircraft Electronic Equipment
MIL-N-18307	- Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-P-23408	- Plating, Tin-cadmium (Electrodeposited)
MIL-C-25050	- Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-L-25467	- Lighting, Integral, Aircraft Instrument, General Specification for

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MIL-L-27160 - Lighting, Instrument, Integral, White General Specification for

MIL-B-27497 - Bearing, Jewel, Sapphire or Ruby Synthetic

MIL-G-81704 - Glass, Aircraft Instruments, Lighting Wedges and Cover

STANDARDS

Federal

FED-STD-595 - Colors

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-143 - Specifications and Standards, Order of Precedence for the Selection of

MIL-STD-147 - Palletized Unit Loads (40" x 48" 4 Way Partial and 4 Way Pallets)

MIL-STD-454 - Standard General Requirements for Electronic Equipment

MIL-STD-471 - Maintainability Demonstration

MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of

MIL-STD-749 - Preparation and Submission of Data for approval of Non-Standard Electronic Parts

MIL-STD-781 - Reliability Tests, Exponential Distribution

MIL-STD-794 - Parts and Equipment, Procedure for Packaging and Packing of

MIL-STD-810 - Environmental Test Methods

MIL-STD-859 - Standard Calibration Table for Aeronautical Pressure Measuring Equipment

MIL-STD-1524 - Table of Differential Pressure in Relation to Calibrated Airspeed

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DRAWINGS

MS27560	- 2 3/8 Square Instrument, Standard Dimension for
MS33558	- Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	- Pointer, Dial Standard
MS33638	- Case, Instrument, Flange Mounted, Aircraft
MS33649	- Bosses, Fluid Connection, Internal Straight Thread

2.1.2 Other Government publications. The following Government publication forms a part of this specification to the extent specified herein.

Department of Defense Standardization

SD-6	- Provisions Governing Qualification

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.4).

3.2 Parts and materials.

3.2.1 General. Parts and materials shall meet the requirements of MIL-E-5400 for Class 2 equipment.

3.2.1.1 Nonstandard parts. Standard MS and AN parts shall be used where they suit the purpose. When no standard part is available, a nonstandard part may be used with prior approval of the procuring activity (see 6.2.1). Selection and approval of the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-STD-749.

3.2.1.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.2 Materials.

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3.2.2.1 Metals. Metals shall be of the corrosion-resistant type unless suitably protected to resist corrosion during the life of the instrument.

3.2.2.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.2.2.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity. (See 6.2).

3.2.2.4 Dissimilar metals. Unless protected against electrolytic corrosion by means of a protective coating, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-454, requirement 16.

3.2.2.5 Protective treatment. When materials which are used in the construction of the airspeed indicator are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.2.6 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity (see 6.2).

3.2.2.7 Toxic and corrosive fumes. The materials, as installed in the airspeed indicator and under the conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.2.2.8 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, case inserts, and fine pitch gear teeth need not be anodized. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.2.2.9 Components. The electronic and mechanical components shall be in accordance with MIL-E-5400.

3.2.2.10 Jewel bearings. All sapphire and ruby jewel bearings used in the airspeed indicator shall be in accordance with MIL-B-27497.

3.2.2.11 Threads. The threads of all machine screws shall be in accordance with MIL-S-7742.

3.3 Design and construction. The indicator shall conform with all applicable requirements of MIL-E-5400 concerning design, construction, and workmanship. The design shall include minimum size and weight consistent with maximum reliability and operability. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the requirements of this specification and any applicable acceptable test documents.

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3.3.1 Physical characteristics.

3.3.1.1 Description. Operation of the indicator shall be accomplished mechanically by means of a precision mechanism which drives a pointer(s), applicable to type, to indicate airspeed on a fixed dial. An integral vibrator shall not be necessary to meet the performance requirements as specified herein. The indicator shall be integrally lighted. All principle dimensions and locations of particular features shall be detailed in table I and figures 1, 2, and 3 for indicator types I, II, and III, respectively.

3.3.1.2 Case. The airspeed indicator case shall conform to MS33638 or MS27560 as applicable to type except as noted herein. The outline dimensions and the external connections of the indicator case shall conform to figures 1, 2, and 3 for indicator types I, II, and III, respectively. The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture. The body shall be finished in a durable black finish, and the mounting flange shall be finished with a durable lusterless black finish, Color No. 37038 of FED-STD-595. The back of the mounting flange shall be tin plated in accordance with MIL-P-23408 to allow electrical grounding to the aircraft instrument panel for compliance with the requirements of MIL-B-5087.

TABLE I. Weight and dimensions for the indicator

Parameter	Value (maximum)	
Type I Indicator		
Length	4.8375 inches	12.287 centimeters
Width	2.385 inches	6.058 centimeters
Height	2.385 inches	6.058 centimeters
1/ Weight	1.25 pounds	0.568 kilograms
Type II Indicator		
Length	4.938 inches	12.542 centimeters
Width	2.385 inches	6.058 centimeters
Height	2.385 inches	6.058 centimeters
1/ Weight	1.25 pounds	0.568 kilograms
Type III Indicators		
Length	7.090 inches	18.009 centimeters
Width	2.385 inches	6.058 centimeters
Height	2.385 inches	6.058 centimeters
1/ Weight	1.25 pounds	0.568 kilograms

1/ Weight of airspeed indicator is without connecting fittings and mounting screws.

3.3.1.3 Pitot and static pressure ports. The pitot pressure boss shall be in accordance with MS33649-4 and the static pressure boss shall be in accordance with MS33649-6. The pressure bosses shall be located in the general locations shown in figures 1, 2, and 3 for indicator types I, II, and III, respectively. A 150-200 wire mesh filter screen (either stainless steel or monel) shall be installed in the static port to prevent dirt particles from entering the case. The inside of the case around each boss and the electrical receptacle shall be sufficiently strong to prevent damage to the case when high torques are applied to the connections during installation and removal in an aircraft. For storage, shipment, and handling pur-

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poses, a vented plug of acceptable material and design shall be provided for the pressure ports. The plugs shall incorporate a vent consisting of a hole at least 1/16 inch in diameter. The plugs shall also incorporate suitable filtering material to prevent the entry of particles which the manufacturer would consider detrimental to the indicator's performance.

3.3.1.4 Application of pressure. The pitot pressure shall be applied to the inside of the pressure capsule and the static pressure to the interior of the case.

3.3.1.5 Capillary tubing. A sufficient length of approved capillary tubing having a minimum inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the indicator case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the requirements of the damping test specified herein.

3.3.2 Reliability.

3.3.2.1 Reliability program. The contractor shall establish a reliability assurance test program in accordance with MIL-STD-781.

3.3.2.2 Operational stability. The airspeed indicator shall meet the performance requirements of this specification when operated continuously or intermittently, for a period of at least 1000 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.3.2.3 Operating life. The airspeed indicator shall have a minimum operating life of 10,000 hours with reasonable servicing and replacement of parts.

3.3.2.4 Reliability in mean-time-between-failure (MTBF). The airspeed indicator shall have a minimum acceptable MTBF θ, of 1,000 hours as specified in MIL-STD-781.

3.3.3 Cabling and connections.

3.3.3.1 Electrical receptacle. The electrical receptacle, used to interface with aircraft wiring, shall be of the type specified in figures 1, 2, and 3 and shall be connected as specified in figures 1, 2, and 3 for indicators type I, II, and III, respectively. The electrical receptacle shall be protected with a suitable cap for shipping, storage, and handling purposes.

3.3.3.2 Case grounding. The case shall be internally grounded as shown in figures 1, 2, and 3 for indicator types I, II, and III, respectively. The electrical system in the indicator shall not be grounded.

3.3.3.3 Interconnecting cabling. The airspeed indicator shall be capable of satisfactory performance using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded. The indicator shall incorporate an electrical receptacle pin grounded inside the case ad-

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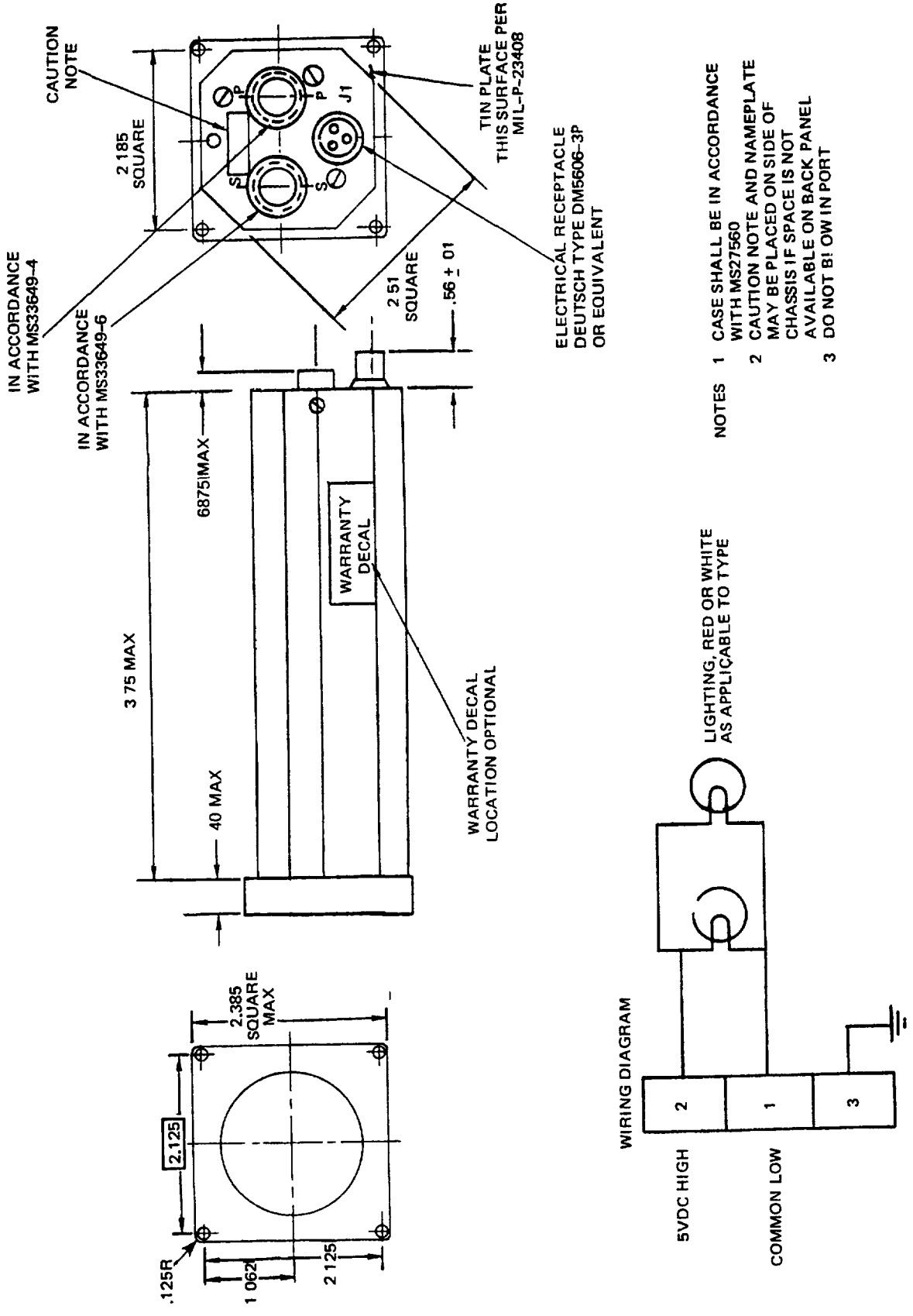
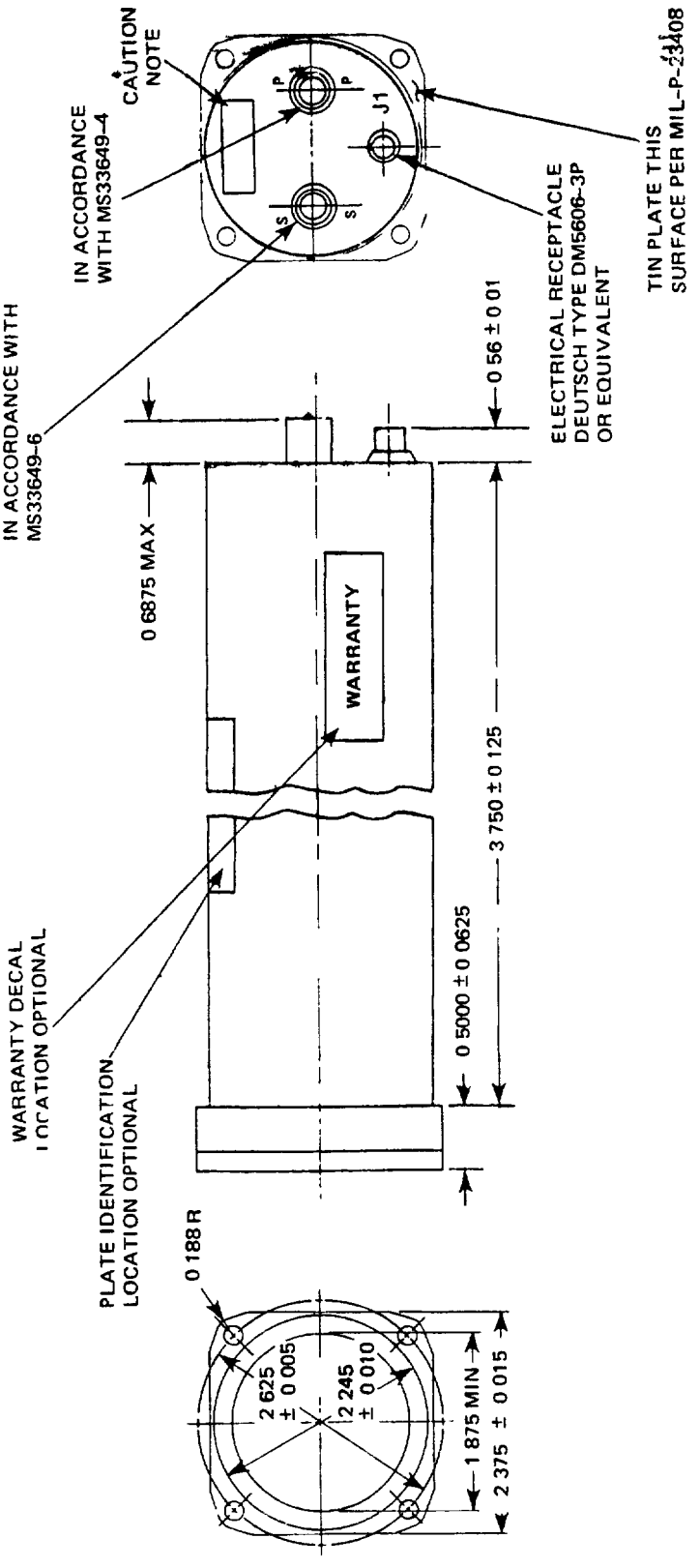
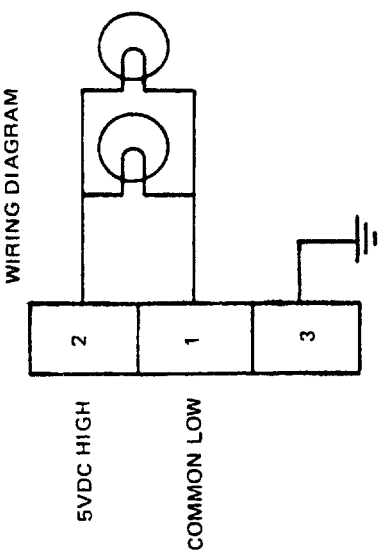


FIGURE 1. Airspeed indicator case, type I.

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NOTES
1 CASE SHALL BE IN ACCORDANCE WITH MS33638
2 DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCE DECIMALS ± 0 005
3 CAUTION NOTE AND NAMEPLATE MAY BE PLACED ON SIDE OF CHASSIS IF SPACE IS NOT AVAILABLE ON BACK PANEL
4 DO NOT BLOW IN PORT.

FIGURE 2. Airspeed indicator case, type II.

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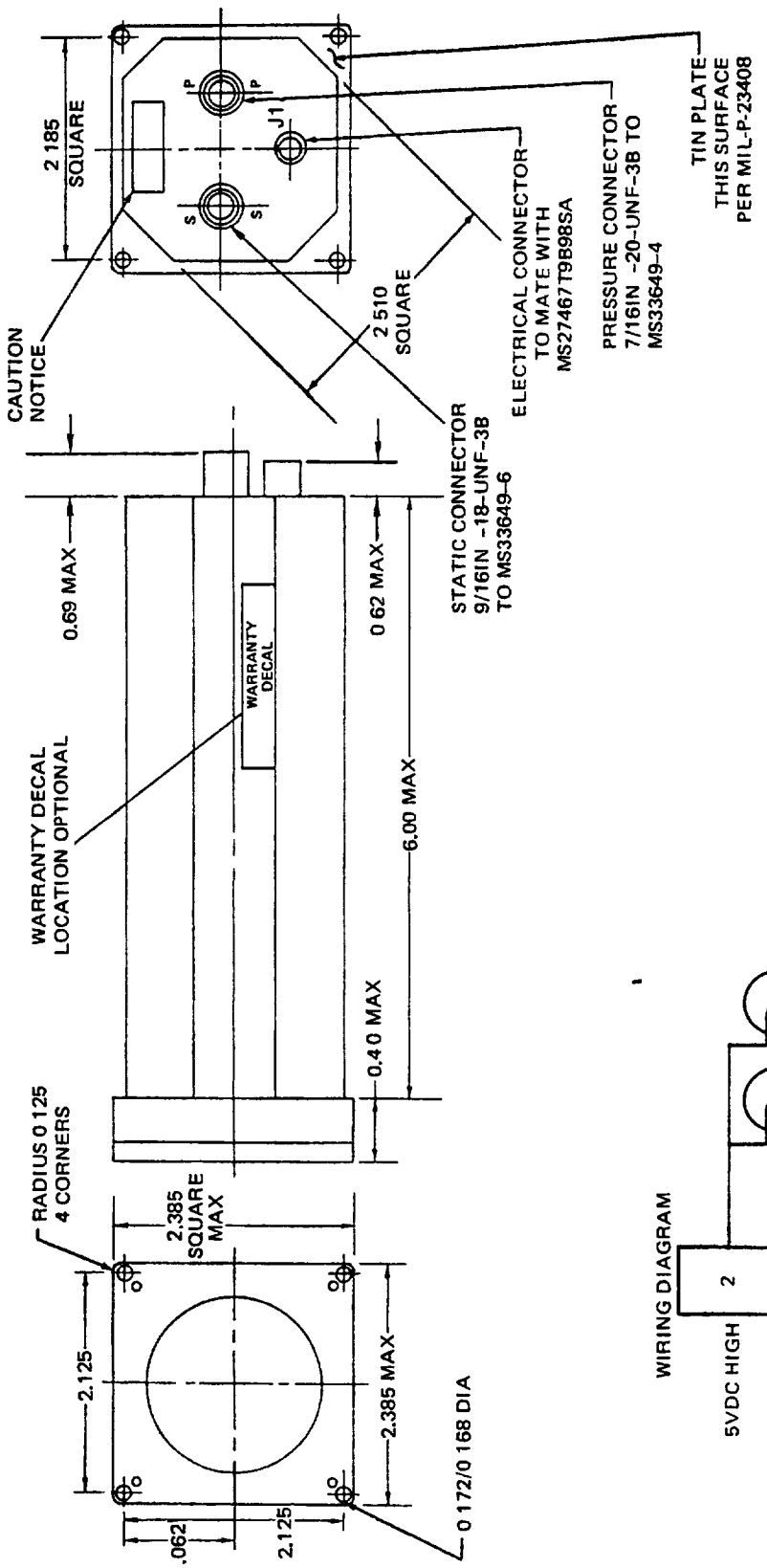


FIGURE 3. Airspeed indicator case, type III.

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jacent to the receptacle as designated in figures 1, 2, and 3. Any changes in wiring must be reported to the qualifying agency as required for retention of qualifications.

3.3.4 Interchangeability. Interchangeability shall exist between all units and replaceable assemblies, subassemblies, and parts for all equipment delivered on the contract in accordance with MIL-STD-454, requirement 7. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000/1.

3.3.5 Interference control. Interference control for the airspeed indicators shall not be required.

3.3.6 Maintainability. The indicator shall be repaired at depot level maintenance shops only. Maintainability shall be demonstrated in accordance with MIL-STD-471. An equivalent plan to demonstrate maintainability prior to the performance of qualification tests may be used upon approval of the qualifying activity (see 6.2.1).

3.3.6.1 Repair time. The airspeed indicator shall be completely repairable at depot level. The maintenance time shall be not greater than 4 hours using one depot technician and authorized depot AGE and technical information. Spare parts and facilities are assumed to be available. Maintenance time shall include all repair and test time excluding government or manufacturer acceptance testing.

3.3.7 Transportability. The airspeed indicator shall be transportable by air, rail, truck, or ship when packaged as specified in section 5.

3.3.8 Shelf life. The indicators shall meet all performance requirements after storage or inactivity for a period of thirty days. The normal storage environmental extremes shall be any dew point between -62°C and +95°C.

3.3.9 Nameplates and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

a. Unit name (Indicated Airspeed Indicator Standby Type I, Type II, Type III)

b. Unit nomenclature

c. Contract number

d. Manufacturer's code number

e. Equipment drawing number

f. Procuring activity (Department of Navy - Naval Air Systems Command)

g. Serial numbers

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3.3.9.1 Decals, nameplates and marking. All decals, nameplates, and markings shall be affixed to the indicator in accordance with figures 1, 2, and 3 for indicator types I, II, and III, respectively and all referenced marking specifications thereon.

3.3.9.2 Serial numbers. Serial numbers will be assigned by the procuring activity for equipment under contract or order. The contractor shall serialize sequentially all equipments which are delivered under the contract.

3.3.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (25 ±5°C)
b.	Pressure	Normal atmosphere (approximately 29.92 in. Hg.)
c.	Vibration	None
d.	Humidity	Room ambient 40 to 90 percent relative humdity
e.	Power	5.0 ±0.1 volt AC or DC (for lighting)

3.3.11 Service conditions. The indicator shall meet the specified operating requirements under the following environmental service conditions.

3.3.11.1 Temperature.

3.3.11.1.1 High temperature. The indicator shall meet the requirements of this specification during and following high temperature exposure in accordance with MIL-STD-810, method 501.1, procedure I except the maximum temperature shall be +71°C (operating) and +95°C (nonoperating).

3.3.11.1.2 Low temperature. The indicator shall meet the requirements of this specification during and following low temperature exposure in accordance with MIL-STD-810, method 502.1, procedure I except the lowest temperature shall be -54°C (operating) and -62°C (nonoperating).

3.3.11.2 Vibration. The indicator shall meet the operational requirements of this specification while being subjected to non-gunfire and gunfire vibration exposure in accordance with MIL-STD-810, method 514.2 and method 519.2, respectively.

3.3.11.3 Acceleration. The indicator shall meet the operational requirements of this specification while being subjected to the acceleration tests for carrier based aircraft in accordance with MIL-STD-810, method 513.2 procedure II. The indicator shall not be physically damaged when subjected to acceleration tests for carrier based aircraft in accordance with MIL-STD-810, procedure I for structural tests.

3.3.11.4 Humidity. The indicator shall operate satisfactorily after being subjected to a humidity test in accordance with MIL-STD-810, method 507.1, procedure I. The indicator shall not be damaged as a result of the humidity test.

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3.3.11.5 Fungus. The indicator shall not be adversely affected by fungi when subjected to fungus tests in accordance with MIL-STD-810, method 508.2.

3.3.11.6 Salt spray. The indicator shall meet the operational requirements of this specification immediately after, and again 48 hours after, salt spray exposure in accordance with MIL-STD-810, method 509.1 procedure I.

3.3.11.7 Rain. The indicator shall meet the operational requirements of this specification after exposure to rain tests in accordance with MIL-STD-810, method 506.1, procedure I and 4.6.18.8.

3.3.11.8 Sand and dust. The indicator shall meet the requirements of this specification when subjected to sand and dust tests in accordance with MIL-STD-810, method 510.1, procedure I.

3.3.11.9 Shock.

3.3.11.9.1 Basic design shock test. The indicator shall meet the operational requirements of this specification during and following basic design shock tests in accordance with MIL-STD-810, method 516.2, procedure I and 4.6.18.10.1.

3.3.11.9.2 Crash safety shock test. The altimeter mounting attachment shall be permitted to bend and distort when subjected to a crash safety shock test in accordance with MIL-STD-810, method 516.2, procedure III and 4.6.18.10.2. However, the mounting attachment shall not break as a result of this test.

3.3.11.10 Dielectric strength. The indicator shall withstand the application of 500 volts ac at a frequency of 60 Hz for 30 seconds.

3.3.11.11 Magnetic property. The indicator shall not cause a free compass to deflect more than 3 degrees when located 5 1/2 inches from the center of the compass.

3.3.11.12 Case leakage. Case leakage shall be not greater than .01 in. Hg. per minute for types I and II indicators and .05 in. Hg. per minute for type III indicator when the pitot and static ports are subjected simultaneously to:

a. A vacuum of 25 in. Hg.

b. A pressure of 10 in Hg.

3.3.11.13 Overpressure. The indicator shall withstand a pressure of 60 in. Hg. applied to the pitot port (73 in. Hg. for type II indicators). The airspeed indicator shall also withstand a pressure of 10 inches of water applied to the static pressure port.

3.3.11.14 Scale error. The indicator shall indicate airspeed within the tolerances of table II.

3.3.11.15 Friction error. The friction error tolerance shall be not greater than the values specified in table III.

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TABLE II. Scale error (knots)

Indicated Airspeed (knots) 1/ Test Points				Differential Pressure	Tolerance ± knots		
	Indicator						
Knots	Type I	Type II	Type III	in. Hg.	Type I	Type II	Type III
20			20	0.0192			2
30	30		30	0.0431	5		2
40	40		40	0.0767	4		2
50				0.120			
60	60	60	60	0.173	3	3	2
70				0.235			
80	80	80	80	0.308	3	3	2
90				0.390			
100	100	100	100	0.481	3	3	3
120	120	120	120	0.695		3	3
140				0.949			
150	150	150	150	1.092	3	3	3
160				1.24			
180				1.58			
200	200	200	200	1.96	5	5	5
220				2.38			
250				3.10			
300	300	300	300	4.53	10	10	10
350				6.29			
400	400	400	400	8.39	15	15	15
450				10.87			
500	500	500	500	13.78	20	20	20
550				17.16			
600	600	600	600	21.08	20	20	20
650				25.63			
700		700		30.76		20	
750				36.57			
800		800		42.94		20	

1/ Refer to MIL-STD-1524

3.3.11.16 Position error. Position error shall be not greater than the values specified in table III.

3.3.12 Service conditions (nonoperating). Service conditions for shelf life and transportation shall meet the shelf life and transit requirements of 3.3.7 and 3.3.8.

3.3.12.1 Warm-up time. Warm-up time for the airspeed indicator shall not be required.

3.3.13 Input electrical power.

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TABLE III. Test tolerances

Test	1/ Indicator Tolerance ±		
	Type I	Type II	Type III
Friction			
20-60 knots	5 knots		6 knots
60-200 knots	4 knots	4 knots	4 knots
200-400 knots	6 knots	10 knots	6 knots
400-600 knots	10 knots	10 knots	10 knots
600-850 knots	-	10 knots	-
Position error			
20-60 knots	4 knots		2 knots
60-120 knots	3 knots	3 knots	3 knots
120-200 knots	3 knots	3 knots	3 knots
200-600 knots	10 knots	10 knots	10 knots
600-850 knots	10 knots	10 knots	-
Pointer Zero Position	2°	10 knots	Pointer indication not greater than 15 knots
Low Temperature Operation (-35°C)			
20-100 knots	4 knots	3 knots	2 knots
100-200 knots	3 knots	3 knots	4 knots
200-850 knots	3 knots	3 knots	5 knots
Low Temperature Operation (-54°C)			
20-150 knots	7 knots	5 knots	3 knots
150-200 knots	5 knots	5 knots	5 knots
200-850 knots	5 knots	5 knots	10 knots
High Temperature Operation (+45°C)	3 knots	3 knots	5 knots
High Temperature Operation (+71°C)			
20-150 knots	5 knots	5 knots	3 knots
150-200 knots	5 knots	5 knots	5 knots
200-850 knots	5 knots	5 knots	10 knots
Overpressure	5 knots	5 knots	5 knots
Damping time required for airspeed pointer to travel from full scale to 120-knot graduation	1 +.1 -.2 seconds		
Vibration Error	4 knots	4 knots	4 knots

1/ Tolerance apply only to points within the range of the indicator.

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3.3.13.1 Operating power. The airspeed indicator shall meet all applicable requirements of MIL-STD-704 and shall give the specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall be not greater than the specified amounts.

a.	Lighting power	5.0 ±0.1 volt, AC or DC, 4.0 watts maximum.

3.4 Performance. The performance of the airspeed indicator shall be as set forth below. Unless otherwise specified herein, these requirements shall apply under both standard and service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

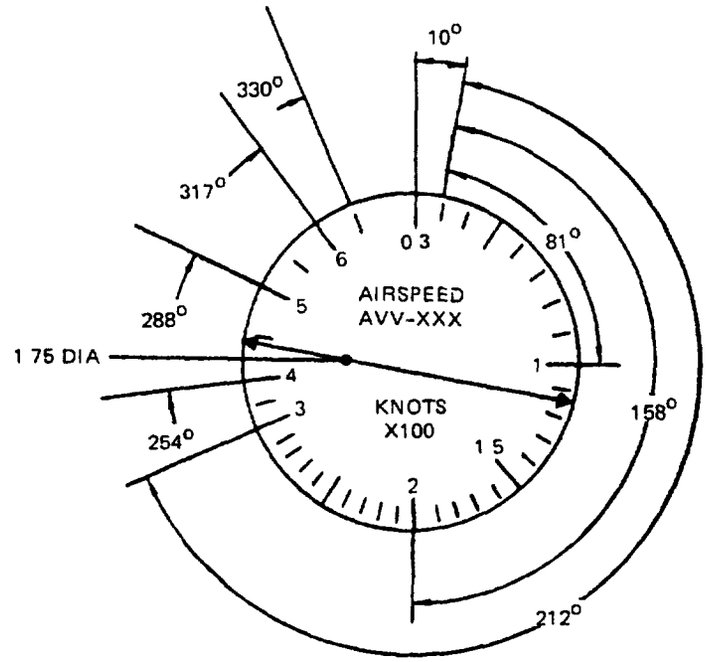
3.4.1 General. The indicators shall be designed to provide indicated airspeed as a function of the difference between pitot and static pressures. The mechanical operation of the airspeed indicator is by a precision mechanism which shall drive a pointer(s) (applicable to type) to indicate airspeed on a fixed dial. (See figures 4, 5, and 6.) The type III indicator shall be capable of providing a display of airspeed by means of two pointers as shown in figure 6. One pointer shall display airspeed in the range of 20-100 knots. Suitable stops shall be provided to limit rotation of this pointer below the zero position or significantly beyond the 100 knot position. The pointer shall be out of view when the 100 knot range is exceeded and shall remain out of view at all times when the indicator is operated beyond this range. A second pointer shall be incorporated to display airspeed over the range of 100 to 600 knots. The tip of the high range pointer shall be out of view until an airspeed of approximately 80 knots is attained. This pointer shall be in full view when the airspeed is 95 knots and shall remain in view throughout the upper operating range of airspeed. The low speed pointer shall indicate no more than 15 knots when no airspeed pressure is applied.

3.4.2 Range. The indicator shall be capable of operating satisfactorily over the range of indicated airspeed as applicable to type (see 1.2). Performance shall be within the requirements of this specification when the altitude is changed at rates of 50,000 feet per minute (FPM) and airspeed is changed at rates of 500 knots per minute.

3.4.3 Integral lighting. The airspeed indicator shall be provided with integral lighting in accordance with MIL-L-25467 and MIL-L-27160, as applicable, to illuminate the display. The lighting circuit shall operate from 5 volts ac or dc and shall be connected as shown in figures 1, 2, and 3. The integral lighting wiring shall not be grounded to the indicator case. The bulbs shall be readily replaceable without soldering unless it can be demonstrated by documentation and test that the lighting system has a minimum MTBF of 10,000 hours. The lighting colors of the airspeed indicator shall be as specified in MIL-C-25050 and in 1.2 of this specification. The design shall be such that red lighting or white lighting, as applicable, can be incorporated by means of a lamp filter change or lamp assembly change only.

3.4.3.1 Coverglass and lighting wedge. The coverglass and/or lighting wedges shall conform to MIL-G-81704 type I and shall not interfere with the readibility of the airspeed indicator. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806.

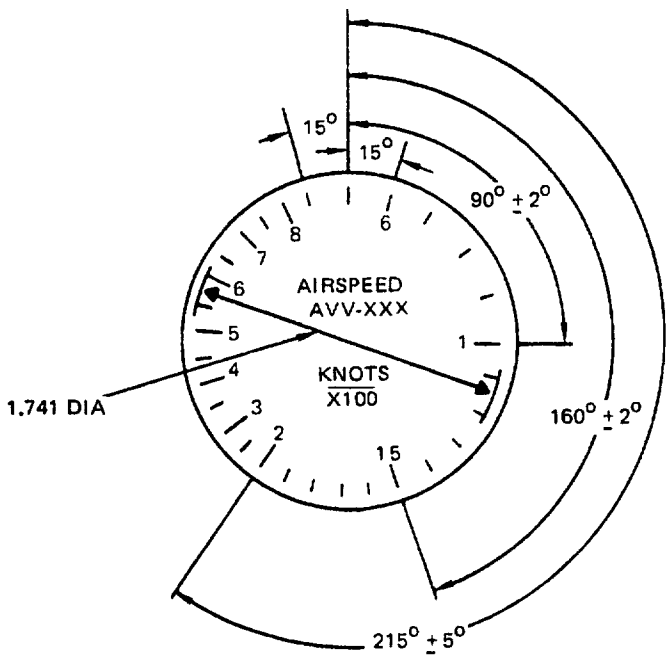
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NOTES 1 DIMENSIONS ARE IN INCHES
2 TOLERANCES ARE DECIMALS ±0 010 AND DEGREES ±1 0

FIGURE 4. Airspeed indicator dial, type I

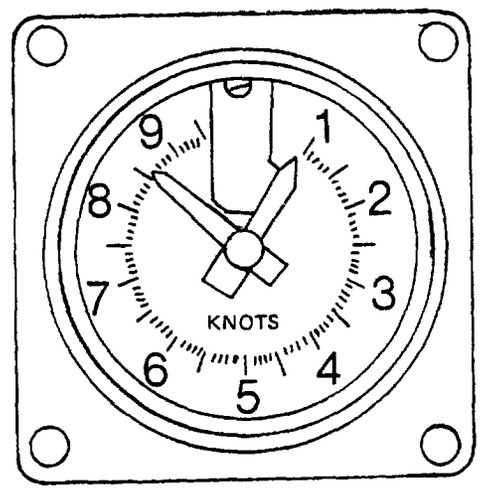
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NOTES 1. DIMENSIONS ARE IN INCHES
2. TOLERANCES ARE DECIMALS ±0.010 AND DEGREES ± 1 0

FIGURE 5. Airspeed indicator dial, type II

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Note: Scale linearity shall be within ± 5%.

FIGURE 6. Airspeed indicator dial, type III

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3.4.3.2 Distance from dial to coverglass. The distance from the dial to the innermost glass surface shall be not greater than 0.250 inch.

3.5 Pointer. The pointer(s) shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. The pointer shall be firmly attached to the pointer shaft to prevent slippage under normal service use and still be removable for depot service maintenance. The pointer shall not distort and bind as a result of application of temperature differentials such as could result from sunlight on the dial or other service conditions. The pointer(s) shall conform to MS33585 and shall be finished in dull white, Color No. 37875 of FED-STD-595.

3.5.1 Pointer movement limits. The airspeed pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 358° of arc clockwise from the no-pressure position when the indicator is subjected to the overpressure tests. The design of the stops shall be approved by the procuring activity. The airspeed pointer movement of type III indicators shall be capable of providing the display described in 3.4.1.

3.5.2 Removal and service. The indicator shall be designed to require no corrective or preventive maintenance while installed in an aircraft.

3.6 Dial. The dial shall be marked as shown in figures 4, 5, and 6 as applicable to type. Unless otherwise specified in the contract or purchase order, all visible portions on the dial shall be lusterless black Color No. 37038 of FED-STD-595, and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. The dimensions of the dial markings shall be as specified in 3.7. The dial shall be securely fastened to the mechanism frame by means of screws or mechanical locking and shall be of sufficient thickness so that warping does not result when the indicator is subjected to the environmental conditions specified herein. The dial shall be removed at overhaul without damage.

3.6.1 Visibility of dial. The pointer, numerals (at least 1/16 inch of the shortest graduations), and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 20 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.2 Dial markings. All dial markings shall be durable so as to withstand the usage encountered in service. The style and proportions of numbers and letters placed on the dial shall conform to MS33558.

3.6.3 Numbers. Each number shall distinctly indicate the graduation to which it applies. If practicable, each number shall be so placed that the center mass of the number is on the radial line joining the appropriate graduation and the center of the dial.

3.7 Finish and marking dimensions.

3.7.1 White marking. The indicator lettering in tables IV and V shall be finished in lusterless white, Color No. 37875 of FED-STD-595. The dimensions shall be as specified in table IV and V.

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TABLE IV. Type I and II indicators marking dimensions (white)

Markings	Height or Length Inch ±0.015	Width of Line or Graduation Inch ±0.005
Numerals 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ON DIAL	0.187	0.023
100 knot graduations	0.171	0.020
50 knot graduations	0.156	0.015
10 knot graduations	0.140	0.015
Lettering knots X100	0.093	0.025
Shaded portion of pointer	-	-

TABLE V. Type III indicators marking dimensions (white)

Markings	Height or Length Inch ±0.015	Width of Line or Graduation Inch ±0.005
Numerals 1, 2, 3, 4, 5, 6, 7, 8, 9	0.190	0.027
KNOTS lettering	0.090	0.015
Major graduations*	0.125	0.020
Minor graduations	0.036	0.010
*except at "5"	0.090	0.020
Shaded portion of pointer	-	-

3.7.2 Back of case marking. The back of the indicator case shall be marked as follows:

a. The letter "P" shall be marked both above and below the pitot pressure port as shown in the applicable figure.

b. The letter "S" shall be marked both above and below the static pressure port as shown in the applicable figure.

c. The electrical receptacle shall be marked with the letter "J1".

3.7.3 Black marking. The indicator lettering in table VI shall be finished in lusterless black Color No. 37038 of FED-STD-595. The dimensions shall be specified in table VI.

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TABLE VI. Type I, II, and III indicators marking dimensions (black)

Markings	Weight or Length Inch ±0.015	Width of Line Inch ±0.005
AVV-XXX, Airspeed Unshaded portion of pointer, dial	0.047 -	0.0156 -

3.8 Workmanship. Workmanship for the indicator shall conform with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that tests and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspections are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Standard conditions. Unless otherwise specified herein, all tests required by this specification shall be made under conditions specified in 3.3.10.

4.3.2 Test readings. The lighting power shall be applied to the airspeed indicator during all tests. The indicator shall be lightly tapped or vibrated before a test reading is taken, unless otherwise specified herein. Vibration shall be applied in the axis perpendicular to the dial at a frequency of 20-60 Hz. at an acceleration level of about 0.2 g.

4.3.3 Attitude. Unless otherwise specified, the airspeed indicator shall be tested in its normal operating position.

4.3.4 Integral lighting. The airspeed indicator lighting system shall be energized and shall operate satisfactorily during all tests when an operational requirement for the airspeed indicator is specified. The lighting shall be tested and operate satisfactorily after completion of each non-operational test.

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4.3.5 Rate of pressure change. Unless otherwise specified herein, the rate of change in pressure during all tests for both increasing and decreasing airspeed and altitude shall be accomplished at any rate up to 500 knots per minute and 20,000 feet per minute, respectively. The rate shall be progressively reduced as the checkpoints are approached to avoid passing the checkpoint. The movement of the pointer during increasing and decreasing airspeed shall be smooth and free from irregular motion.

4.3.6 Standard pressure. The standard pressures used in calibrating and testing the airspeed indicator shall be as specified in MIL-STD-1524 for differential pressure in relation to calibrated airspeed and MIL-STD-859 for static pressure.

4.4 Qualification inspection. Qualification inspection on sample units shall be performed at a laboratory acceptable to the Government (see 6.4). These sample units should be manufactured by normal production procedures.

4.4.1 Sample. Three airspeed indicators shall be subjected to the qualification inspection. The samples shall be taken at random from a production run. The sample indicators shall have been subjected to and passed the requirements of group A inspection (see 4.5.1.2). Qualification shall not be granted if group A inspections are not met. Each indicator type (I, II, and III) shall be qualified separately.

4.4.2 Inspection routine. Sample indicators shall be subjected to the qualification inspection specified in table VII in the order shown.

4.4.2.1 Qualification inspection sample identification. The qualification test samples shall be plainly identified by durable tags, securely attached and marked with the following information:

Qualification Test Sample

Airspeed Indicator Type _____. Submitted by (manufacturer's name, date). For qualification inspection in accordance with MIL-I-85506(AS) under authorization (reference letter authorizing tests).

4.4.3 Failures. Failure to meet test requirements shall be cause for refusal to grant qualification.

4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report concerning production units at 2-year intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B), indicating as a minimum the number of lots that have passed, the number that have failed, and the group which they failed. The results of tests of all reworked lots shall be identified and accounted for.

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b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 2-year period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 2-year period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 2-year period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Inspection lot. A normal lot size shall be 25 indicators. Refer to table VIII. If the lot size is less than 25 indicators, the quantity produced in two months shall be considered one lot. Unless otherwise specified in the contract, concurrent delivery shall be permitted during the storage period. When an indicator failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of the failure has been determined and appropriately corrected. When a failure occurs, shift to one sample out of each 15 from those still on hand and proceed with sampling, as indicated in table VIII, when the next quantity is offered for acceptance. In the event that a failure is noted for the sample which is subjected to the shelf life test only, the shelf life test is the sole test to which the increased sample must be subjected.

4.5.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table IX. These inspections should be performed in the order shown, except as noted.

4.5.1.2.1 Sampling plan. Subgroup 1 tests shall be performed on every indicator supplied under this specification. Airspeed indicators that do not meet the requirements of the subgroup 1 shall be removed from the lot. Lots having more than 20 percent total rejects shall not be furnished on contracts. Subgroup II samples shall be selected from production at random in accordance with the table VIII. One additional sample, selected at random, shall also be tested from each monthly lot for shelf life only in accordance with 4.6.15.

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TABLE VII. Qualification inspection.

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of failures allowed 1/
Examination of Products	3	3.2. through 3.3.1.5	4.6.1	1/
Manufacturing Run-in			4.6.2	
Case Leakage		3.3.11.12	4.6.3	
Differential Pressure Capsule Leakage			4.6.4	
Airspeed Display Scale Error		3.3.11.14	4.6.5	
Friction error		3.3.11.15	4.6.6	
Position error		3.3.11.16	4.6.7	
Dielectric Strength		3.3.11.10	4.6.9	
Scale Error at -35°C		3.3.11.14	4.6.10	
Scale Error at +45°C		3.3.11.14	4.6.11	
Vibration Error			4.6.12	
Damping			4.6.13	
Overpressure		3.3.11.13	4.6.14	
Shelf Life		3.3.8	4.6.15	
Magnetic Property		3.3.11.11	4.6.17	
Low temperature Exposure		3.3.11.1.2	4.6.18.1	
High temperature Exposure		3.3.11.1.1	4.6.18.2	
Vibration		3.3.11.2	4.6.18.3	
Acceleration		3.3.11.3	4.6.18.4	
Humidity		3.3.11.4	4.6.18.5	
Fungus		3.3.11.5	4.6.18.6	
Salt Spray		3.3.11.6	4.6.18.7	
Rain		3.3.11.7	4.6.18.8	
Sand and Dust		3.3.11.8	4.6.18.9	
Impact Shock		3.3.11.9	4.6.18.10	
Lighting		3.4.3	4.6.19	
Internal Examination		3.3, 3.3.1	4.6.20	
Maintainability		3.3.6	4.6.21	
Reliability Qualification Phase		3.3.2	4.6.22.1	

1/ No failures shall be permitted. Any discrepancies, failures, or unacceptable performance must be evaluated by the procurement/qualifying agency to determine if corrective action can be taken without repetition of any tests.

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TABLE VIII. Group A inspection, subgroup II, sample selection.

Quantity Offered for Acceptance	Quantity to be Selected for Tests
First 15	1 (0 for B Inspection)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

4.5.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup I, table IX as the final step of his production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1 tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests (see 6.2.2).

f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.5.1.2.1.2 Rejected lots (subgroup 2). Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100 percent inspection on networks of the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected using the tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed.

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4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table X. They shall be performed on sample units that have successfully passed group A inspections. Group B inspection shall be performed in the order shown, except as noted.

TABLE IX. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	Sampling procedures
Subgroup I			
Examination of Product	3.2. through 3.3.1.5	4.6.1	
Manufacturing Run-in		4.6.2	
Case Leakage	3.3.11.12	4.6.3	
Differential Pressure Capsule Leakage 1/		4.6.4	All sample units
Airspeed Display Scale Error 1/	3.3.11.14	4.6.5	
Friction error 1/	3.3.11.15	4.6.6	
Position error	3.3.11.16	4.6.7	
Lighting	3.4.3	4.6.8	
Subgroup II (Individual tests)			
Dielectric Strength 1/	3.3.11.10	4.6.9	
Scale Error at -35°C 1/	3.3.11.14	4.6.10	
Scale Error at +45°C 1/	3.3.11.14	4.6.11	
Vibration Error 1/		4.6.12	(See 4.5.1.2.1)
Damping		4.6.13	
Overpressure 1/	3.3.11.13	4.6.14	
Shelf Life	3.3.8	4.6.15	
Lighting	3.4.3	4.6.16	

1/ These tests may be performed in any order.

4.5.1.3.1 Sampling plan. Three production samples shall be selected at random from the first 15 produced on contract and submitted within 10 days after being manufactured. Unless otherwise directed by contract or purchase order, these samples shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Airspeed Indicator, Type ____. Submitted by (manufacturer's name, date) for production Group B Inspection of MIL-I-85506(AS) with Contract No. _____. Manufacturer's part number _____.

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Along with the test sample, the manufacturer shall provide a letter of certification which states therein that the test samples do not differ from the currently qualified design in any detail such as materials, parts, or assembly. When any details or design changes have been incorporate or have altered the qualified design in anyway, the manufacturer shall submit a complete disclosure of the changes made. When such changes have affected the drawings, parts lists, or other documentation submitted to the government for qualifications, the manufacturer shall submit updated documentation as necessary.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered per the contract or purchase order, provided they are not scheduled for group C inspections and meet requirements for visual and mechanical inspection.

TABLE X. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of failures Allowed
Magnetic Property 3/	3.3.11.11	4.6.17	2/
Low Temperature 3/	3.3.11.1.2	4.6.18.1	
High Temperature 3/	3.3.11.1.1	4.6.18.2	
Vibration 3/	3.3.11.2	4.6.18.3	
Acceleration 3/	3.3.11.3	4.6.18.4	
Humidity	3.3.11.4	4.6.18.5	
Fungus	3.3.11.5	4.6.18.6	
Salt Spray	3.3.11.6	4.6.18.7	
Rain	3.3.11.7	4.6.18.8	
Sand and Dust	3.3.11.8	4.6.18.9	
Impact Shock	3.3.11.9	4.6.18.10	
Lighting	3.4.3	4.6.19	
1/ Internal Examination	3.3, 3.3.1	4.6.20	
Maintainability Demonstration	3.3.6	4.6.21	

1/ The airspeed indicator selected for internal examination shall not be exposed to the fungus test.

2/ No failures shall be permitted. Any discrepancies, failures, or unacceptable performance must be evaluated by the procurement/qualifying agency to determine if corrective action can be taken without repetition of any tests.

3/ These tests may be performed in any order.

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4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.5), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table XI, in the order shown. Group C inspection shall be performed on sample units of each style and selected from inspection lots which have passed groups A and B inspections. Group C inspection samples shall be representative of production.

4.5.2.1.1 Sampling plan. Three (3) to six (6) airspeed indicators, selected at random, shall be tested from the monthly lot for accept-reject criteria of test plan IV. If the lot size is less than 25 indicators, the quantity produced in two months shall be considered as one lot. At the conclusion of each month, the test results shall be made available for the procuring activity (see 6.2.2). At any time that the test results show a reject situation, the procuring activity shall be notified (see 6.2.2).

4.5.2.1.2 Failure criteria. If recalibration, adjustment, or repair is required to pass any of the tests listed in 4.6.22.4, either at the 250 hour test intervals or at the completion of the reliability test, a failure shall be charged to that test. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one part caused the other parts to fail.

4.5.2.1.3 Preventive maintenance. No preventive maintenance shall be performed on the airspeed indicator during the 1,000 hour MTBF test. The failures occurring during the longevity test shall be analyzed in accordance with MIL-STD-781. In addition, the analysis of failures, especially for what appears to be pattern type failures, should indicate whether preventive maintenance would have prevented the failure and extended the service life.

TABLE XI. Group C inspection.

Test or Inspection	Requirement Paragraph	Test Method Paragraph	Number of Sample Units for Inspection
Reliability Sampling Phase Test	3.3.2	4.6.22.2	3-6 units per month
Longevity Tests	-	4.6.22.3	2 units

4.5.2.1.4 Disposition of samples. Any reliability sample tested may be delivered on contract provided the sample is in "good as new" condition or has been refurbished and meets the group A, subgroup I inspections and is representative of production units currently being accepted.

4.5.2.1.5 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure (see 6.2.2) and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and

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which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity (see 6.2).

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

4.6 Methods of inspection. The procedures used for conducting qualification and quality conformance tests shall be prepared by the contractor in accordance with MIL-T-18303 and the following paragraphs for procuring activity review and approval (see 6.2.2). The right is reserved by the procuring activity or the Government inspector to modify the tests or require addition tests deemed necessary to determine compliance with the requirements of this specification or the contract.

4.6.1 Examination of product.

4.6.1.1 Qualification inspections. Prior to final assembly, the airspeed indicators that are to be subjected to qualification inspections shall be examined to verify compliance with each of the design requirements of section 3 not covered by specific tests.

4.6.1.2 Quality conformance inspection. Each airspeed indicator selected for quality conformanace inspection shall be examined externally to determine conformance with the applicable drawings and with each of the requirements of this specification capable of being inspected in this manner.

4.6.2 Manufacturing run-in. Each airspeed indicator shall be subjected to a manufacturing run-in test of six hours duration. The first two hours of the run-in shall be accomplished with the airspeed indicator exposed to a temperature of +71°C. The remaining four hours of the test shall be accomplished under room ambient conditions with the airspeed indicator mounted on a special test table. The table shall subject the indicator to roll, pitch, and yaw motion (Scorsby) at a frequency of five to seven oscillations per minute through an angle of 30 degrees. Prior to initiation of the run-in test, the scale error shall be determined and recorded. The scale error shall be within the room temperature scale error tolerance of table II. During the performance of the entire run-in tests, the indicator shall be subjected to vibration of approximately 1/2 g. as measured on the instrument panel at frequencies between 20 and 60 Hz. The 5 volt lighting circuit shall be energized continuously during the test. The airspeed indicator shall be pneumatically cycled during the six hour run-in test. The pneumatic cycle shall range from ambient (zero differential) pressure to about 500 knots and return to ambient within approximately 15 minutes for a total of at least 4 cycles per hour. Failures noted for production airspeed indicators during run-in tests shall be summarized for the monthly production reliability report (see 6.2).

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4.6.3 Case leakage. The pitot and static pressure connections of the indicator shall be joined with a Y connection to a mercury manometer and a source of vacuum and pressure. A vacuum of 25 in. Hg. shall be applied to the pitot and static connections of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. This procedure shall be repeated with a pressure of 10 in. Hg. In each case, during a period of 1 minute, the mercury level shall not change more than .01 in. Hg. for type I and II indicators and .05 in. Hg. for type III indicator.

4.6.4 Differential pressure capsule leakage. With the static connection open to the atmosphere, a pressure sufficient to produce a full scale deflection of the pointer shall be applied to the pitot connection of the indicator. The connecting tubing shall then be pinched off or otherwise completely sealed. During a period of one minute, the indicator pointer shall not change its position by more than one knot. For type III indicators, the indicator pointer shall not change its position by more than 5 knots. This leakage rate shall be confirmed by the use of a manometer.

4.6.5 Airspeed display scale error (room temperature). The indicator shall be tested for scale errors at the applicable test points shown in table II. The test shall be performed by subjecting the differential pressures specified to produce these readings, first with increasing differential pressures, then with decreasing differential pressures. For increasing differential pressures, the pressure shall be brought up to but shall not exceed the differential pressures required to produce the required setting and for decreasing differential pressures, the pressure shall be brought down to but not below the differential pressures required to produce the required setting. The indicator errors shall not exceed the tolerances specified in table II for the indicator type being tested.

4.6.6 Friction. The indicator shall be tested for friction at the scale error test points applicable to the indicator being tested as listed in table III. The differential pressure shall be increased to bring the pointer to the approximate desired reading and held constant while two readings are being taken; the first before the indicator is tapped or vibrated, the second after the indicator friction is removed. The difference between any two readings shall not exceed the values specified in table III. The pointer shall move smoothly while the pressure is varied uniformly without vibrating the indicator. The test may be combined with the scale error test at room temperature.

4.6.7 Position error. The indicator shall be in the normal mounting position with no differential pressure applied. The deviation of the pointer from the zero graduation shall be within the tolerance supplied in table III. Type III indicators shall indicate no more than 15 knots in this position. Differential pressure shall then be applied as required to obtain a reading of approximately 100 knots (60 knots for type III indicators). The pressure shall be held constant until a reading is obtained. The indicator shall then be rotated in such a manner as to position the zero graduation at the 90°, 180° and 270° positions of rotation, respectively, and a reading shall be taken at each position. A reading shall also be taken with the indicator dial face up (horizontal). The difference in readings taken for the normal position and any other position shall not be greater than the tolerance specified in table III. The indicator shall be lightly tapped or vibrated during this

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test. The test shall be repeated for differential pressures 200 knots and 600 knots during qualification inspections and sampling inspections.

4.6.8. Lighting (group A inspection). Lighting tests specified as Sampling Plan A tests in MIL-L-25467 and MIL-L-27160 shall be performed as applicable. Those tests which are common may be performed once.

4.6.9 Dielectric strength. A voltage of 500 volts shall be applied from receptacle pins 1 and 2 (together) to ground for a minimum of 30 seconds. The voltage shall be applied by means of a rheostat control and raised from zero to maximum. There shall be no insulation breakdown or permanent damage to the indicator as a result of this test.

4.6.10 Low temperature operation (-35°C). The indicator shall be subjected to a temperature of -35 ±2° for a period of 4 hours. At the end of this period and while still at this temperature, the integral lights of the airspeed indicator shall be turned on and a scale error test shall be performed. Upon completion of the scale error test, the indicator shall be tested for case leakage (see 4.6.3). The difference in scale errors shall not be greater than the applicable tolerances of table III. The integral lighting shall operate satisfactorily during this test. No damage to the indicator shall result from the exposure to low temperatures and case leakage requirements shall be met.

4.6.11 High temperature operation (+45°C). The indicator shall be placed in a chamber and maintained at a temperature of +45 ±2°C for a period of 4 hours. At the end of this period and while still at this temperature, the integral lights of the airspeed indicator shall be turned on and a scale error test shall be performed. Upon completion of the scale error test, the indicator shall be tested for case leakage (see 4.6.3). The difference in scale errors shall not be greater than the applicable tolerances of table III. The integral lighting shall operate satisfactorily during this test. No damage to the indicator shall result from exposure to high temperatures and case leakage requirements shall be met.

4.6.12 Vibration error. The indicator shall be mounted on a vibration stand in its normal operating position and secured by flange screws positioned as on an aircraft panel. The indicator shall be subjected to one hour of vibration frequencies from 500 to 3,000 cpm with an amplitude between 0.009 and 0.011 inch. The vibration shall be such that any point on the indicator case will describe the circular path of amplitude specified. The circular path shall be inclined at an angle of 45° to the horizontal and vertical (dial) planes. The frequency shall be varied uniformly from 500 cpm to 3,000 cpm and return to 500 cpm during the hour. A differential pressure of 200 knots (60 knots for type III indicators) shall be applied to the airspeed indicator during this test. The natural frequency of the airspeed indicator shall not occur in the range of the test. While the indicator is being vibrated, the maximum spread of pointer vibration and pointer variation from its original position shall not exceed the tolerances specified in table III. Subsequent performance of the indicator upon completion of the vibration test shall not be affected.

4.6.12.1 Alternate vibration error. An equivalent vibration test using a linear shaker may be performed as an alternative to the preceding test. The airspeed indicator shall be vibrated in three mutually perpendicular directions at the appropriate amplitudes. The test procedures shall be in accordance with MIL-STD-810 and shall be approved by the procuring or qualifying activity.

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4.6.13 Damping. Sufficient pressure shall be applied to the pitot connector to produce full-scale deflection of the airspeed pointer. The pressure shall be suddenly released. The time required for the airspeed pointer to reach the value specified in table III shall be within the tolerances specified.

4.6.14 Overpressure. A pressure of 60 in. Hg. (73 in. Hg. for type II indicators) shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes. A pressure of 10 inches of water shall be applied to the static pressure connection only of the indicator for a period of 10 minutes. Not less than 5 minutes after the applications of the preceding noted pressures, the indicator shall be tested for room temperature scale errors at the applicable test points in table II. The errors shall not exceed the increase in tolerances specified by table III. The increase in tolerances shall not apply to subsequent scale error tests which are performed after 24 hours has elapsed.

4.6.15 Shelf life. The indicator shall be placed in storage for a period of not less than 30 days. During the 30-day period, the indicator shall remain as dormant as practicable. The mechanism shall not be exercised other than that caused by normal ambient pressure vibrations. The temperature shall remain essentially constant; however, normal room temperature variations will be permissible. After the 30-day period, the indicators shall again be subjected to the following tests:

a. Case leakage (see 4.6.3).

b. Differential pressure capsule leakage (see 4.6.4).

c. Scale error at room temperature (see 4.6.5).

d. Friction (see 4.6.6).

4.6.16 Lighting (group A, subgroup II inspection). The indicator shall be tested to verify that the integral lighting meets the applicable Sampling Plan A requirements of MIL-L-25467 or MIL-L-27160.

4.6.17 Magnetic property. The airspeed indicator shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the indicator 5 1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the airspeed indicator shall be rotated about on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. With the indicator at any specified position, the compass deflection shall be not greater than 3°.

4.6.18 Environmental tests. The airspeed indicator shall be subjected to the following tests to be performed in accordance with the specified procedures of MIL-STD-810. Upon completion of each environmental test, the airspeed indicator shall be subjected to and meet the requirements of the following tests:

a. Case leakage (see 4.6.3).

b. Differential pressure capsule leakage (see 4.6.4).

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c. Airspeed display scale error (see 4.6.5).

d. Friction (see 4.6.6).

e. Integral lighting (see 4.6.8).

When external connections are required to simulate installed conditions, they shall consist of standard AN or MS fittings and at least 10 ft of 1/4 inch or 3/8 inch tubing, coiled and so arranged as to allow drainage of condensed vapors. The tubing shall be attached to the static pressure and pitot pressure ports of the indicator.

4.6.18.1 Low temperature. The low temperature test shall be performed in accordance with MIL-STD-810, method 502.1, procedure I. The operating temperature shall be -54°C. The operation tests at -54°C shall consist of airspeed display scale error test (see 4.6.5). The difference in scale errors shall not be greater than the applicable tolerances of table III.

4.6.18.2 High temperature. The high temperature test shall be performed in accordance with MIL-STD-810, method 501.1, procedure I. The test temperature shall be +95°C. The period of exposure shall be 4 hours. The temperature at which operational tests are to be performed shall be 71°C. The operational tests at 71° shall consist of airspeed display scale error test (see 4.6.5). The difference in scale errors shall not be greater than the applicable tolerances of table III.

4.6.18.3 Vibration. The airspeed indicator shall be subjected to vibration tests in accordance with MIL-STD-810 and tables XII and XIII of this specification. The indicator shall be mounted in the normal operating position for all vibration tests and the integral lighting shall be electrically energized and operating. During vibration resonance searches as required to be performed as part of sinusoidal cycling functional tests, the pressure to the indicator shall be maintained constant at about 200 knots. As the resonant search is performed, the indicator display (pointer) shall not exhibit oscillations greater than 5 knots and pointer variation from the normal shall not exceed 5 knots. Upon completion of a resonance search, the remainder of the sinusoidal cycling functional vibration test for each axis shall be performed while the indicator is being pressure cycled. The pitot pressure to the indicator shall be cycled from room ambient (zero differential) pressure to approximately 3/4 of the airspeed range and return to ambient at a rate of about 100 knots per minute. The functioning of the indicator under test shall be noted to be satisfactory with respect to smoothness of operation (friction). The indicator shall also be pressure cycled during random vibration functional tests and performance (friction) shall be satisfactory. During vibration endurance tests (non-gunfire and gunfire), the indicator shall also be energized and pressure cycled, but functional performance effects on the indicator need not be evaluated. After the completion of functional tests, the indicator shall be subjected to, and meet the requirements of the tests listed in 4.6.18. After completion of all the endurance tests, the indicator shall again be subjected to the test listed in 4.6.18. The results of the tests shall be within the tolerances specified except that an increase of 5 knots is permitted for friction test results. A scale error (bias) shift of 5 knots shall also be permitted.

4.6.18.4 Acceleration. Acceleration tests shall be performed in accordance with MIL-STD-810, method 513.2, procedure I for structural test and procedure II for operational test.

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TABLE XII. Vibration test (Non-gunfire).

Frequency Range (Hz)	Type of Motion	Test Time Per Axis		MIL-STD-810 Applicable Curve	Airspeed Test Requirement
		Sweep Time	Sweep Rate Cycle		
5-50	Sinusoidal Cycling	30 minutes (Includes resonance dwell of 5 minutes)	7.5 minutes	Figure 514.2-2, 5-33 Hz, Curve C; 33-50 Hz, Curve B (2g)	Functional (Performance)
50-2000	Random	30 minutes	20 minutes	Figure 514.2-2A,50-1000 Hz; Maximum acceleration power spectral density Wo^2 = 0.009 G^2/Hz. Minimum overall G's rms=3.6 1000-2000 Hz; -6dB/octave	Functional (Performance)
5-50	Sinusoidal cycling	90 minutes (Includes resonance dwell time 30 minutes)	7.5 minutes	Figure 514.2-2, 5-33 Hz, Curve C; 33-50 Hz, Curve B (2g)	Endurance (Non Functional)
50-2000	Random	60 minutes	20 minutes	Figure 514.2-2A,50-1000 Hz; Maximum acceleration power spectral density Wo=0.027 G^2/Hz. Minimum overall G's rms=6.3 1000-2000 Hz; -6dB/octave	Endurance (Non Functional)

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TABLE XIII. Vibration test (gunfire).

Frequency Range (Hz)	Type of Motion	Test Time Per Axis		MIL-STD-810 Applicable Curve	Airspeed Test Requirement
		Sweep Time	Sweep Rate Cycle		
50-500	Sinusoidal cycling	60 minutes	7.5 minutes	Figure 514.2-2,50-60 Hz; 0.025 inch double amplitude D.A. to ±3.5g 60-220 Hz; ±3.5g to 0.00156 D.A. 220-500 Hz; 0.00156 to ±7g.	Endurance (Non Functional)
500-2000	Random	15 minutes		Figure 514.2-2A,500-1000 Hz; Maximum acceleration power spectral density Wo= $0.067G^2$/Hz; Minimum overall G's rms 7.6 1000-2000 Hz, -9dB/ octave	Endurance (Non Functional)

4.6.18.4.1 Operational test, procedure II. The airspeed indicator shall be subjected to tests on a centrifuge while oriented in each of the six positions defined in MIL-STD-810, method 513.2. The indicator shall be installed so that it can be electrically energized and pneumatically operated during the test. The indicator shall be subjected to the "g" levels listed in table XIV. The test shall be performed with the indicator pitot and static pressure ports exposed to a constant differential pressure of approximately 300 knots. The change in airspeed scale errors recorded for each position during acceleration shall be not greater than 1.5 knots per "g". The indicator lighting shall be noted to be continuous (without flicker) during acceleration. The inspections listed in 4.6.18 shall be performed upon completion of this acceleration test.

4.6.18.4.2 Structural test, procedure I. The airspeed indicator shall be subjected to tests on a centrifuge while oriented in each of the six positions defined in MIL-STD-810, method 513.2. The indicator shall be electrically energized during this test, and a constant differential pressure of approximately 300 knots shall be maintained at the pressure ports. Indicator accuracy need not be recorded during this test. The indicator shall be subjected to the "g" levels listed in table XV. Upon completion of this acceleration test, the indicator shall be subjected to the inspections listed in 4.6.18.

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TABLE XIV. Operational test "g" levels.

MIL-STD-810 Method 513.2 Indicator Position	Centrifuge "g" Test Level	
	Type I and II Indicators	Type III Indicator
Fore	2.0	1.0
Aft	6.0	1.0
Up	9.0	7.6
Down	3.0	5.3
Lateral, right side	4.0	2.2
Lateral, left side	4.0	2.2

TABLE XV. Structural tests "g" levels

MIL-STD-810 Method 513.2 Indicator Position	Centrifuge "g" Test Level
Fore	3.0
Aft	9.0
Up	13.5
Down	4.5
Lateral, right side	6.0
Lateral, left side	6.0

4.6.18.5 Humidity. The humidity test shall be performed in accordance with MIL-STD-810, method 507.1, procedure I except the test item may be removed from the chamber at the completion of the exposure period for the performance of tests. The indicator shall be thoroughly examined within at least 48 hours after completion of the humidity exposure and there shall be no evidence of corrosion and deterioration that would affect subsequent operation of the indicator.

4.6.18.6 Fungus. The fungus test shall be performed in accordance with MIL-STD-810, method 508.2. At the end of test period, the indicator shall be thoroughly examined to ascertain that no fungus growth has occurred. This test need not be performed if a detailed review of all indicator components is made and appropriate certification is provided concerning the use of only non-nutrient material in accordance with MIL-STD-454, requirement 4.

4.6.18.7 Salt spray. The salt spray test shall be performed in accordance with MIL-STD-810, method 509.1, procedure I. The airspeed indicator shall be subjected to the tests listed in 4.6.18 immediately after completion of the salt spray exposure period and again 48 hours (approximately) later.

4.6.18.8 Rain. The rain test shall be performed in accordance with MIL-STD-810, method 506.1, procedure I. Operation of the indicator during exposure to the rain shall not be required. The front surface (dial) only shall be directly exposed to the rain. The test duration shall be for a period of 30 minutes.

4.6.18.9 Sand and dust. The sand and dust test shall be performed in accordance with MIL-STD-810, method 510.1, procedure I.

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4.6.18.10 Shock.

4.6.18.10.1 Basic design. This test shall be performed in accordance with MIL-STD-810, method 516.2, procedure I, figure 516.2-1, peak value (a), time duration (c). The airspeed indicator shall be operated during this test. The integral lighting shall be on and a constant differential pressure of approximately 120 knots shall be applied. Prior to the application of impacts, the airspeed indication shall be recorded. Monitoring of indicator performance during application of impact shocks is not required. Upon completion of the tests for each axis, the airspeed indication should again be recorded and the readings compared to readings taken prior to the application of impacts. The airspeed indicator and the data should be examined to determine that mechanical or electrical failure did not occur. Upon completion of the procedure, the indicator shall be subjected to and shall meet the requirements of inspections listed in 4.6.18.

4.6.18.10.2 Crash safety. This test shall be performed in accordance with MIL-STD-810, method 516.2, procedure III, figure 516.2-1, peak value (a), time duration (c). As specified, a dummy load may be used in lieu of an indicator. Upon completion of the crash safety shock test, the test item shall be examined and all damage recorded. Bending and distortion of the mounting attachment shall be permitted and not counted as a failure. A broken mounting attachment shall be counted as equipment failure. Operation of the indicator after the shock test is not required.

4.6.19 Lighting (group C inspection). Lighting tests shall be conducted in accordance with Sampling Plan B tests of MIL-L-25467 and MIL-L-27160 as specified. Those tests which are common to both specifications may be conducted once. Integral lighting colors shall be in accordance with MIL-C-25050.

4.6.20 Internal examination. Only one indicator shall be subjected to this test. The case of the airspeed indicator shall be opened and the internal mechanism shall be examined to determine conformance with the applicable requirements of the specification. The lamp circuit of the indicator shall also be examined for conformance to the requirements of MIL-L-25467. In addition, any damage or deterioration to the mechanism that resulted from any of the preceding tests, or any other deficiency which could in any manner prevent the indicator from meeting operational or maintenance requirements during service life, shall be cause for rejection. The qualification sample indicator shall be re-cased and demonstrated to perform satisfactorily during abbreviated room temperature tests. Such indicators may be cleaned prior to re-casing. The internal examinations may be combined with the maintainability demonstration (see 4.6.21).

4.6.21 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, test method 1, plan B1. Maintainability demonstration shall be performed on a minimum of three selected repairs. The tasks selected to demonstrated maintainability shall be selected in advance and shall be approved by the procuring activity (see 6.2.2). An equivalent maintainability demonstration may be performed prior to initiation of qualification testing on approval of the qualifying activity. If applicable, interchangeability of lighting (color) may also be demonstrated.

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4.6.22 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781 for the Qualification Phase and Sampling Phase tests. Airspeed indicators selected for reliability assurance tests shall first be subjected to the inspection tests of 4.6.18.

4.6.22.1 Reliability qualification phase. Prior to the acceptance of the airspeed indicators under contract or order, three (3) to six (6) indicators shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests", using test level E, except as specified herein. Criteria for test plan II shall be used. The airspeed indicator shall demonstrate a minimum acceptable MTBF θ_1, of 1000 hours as defined in MIL-STD-781.

4.6.22.1.1 Duty cycle. The duty cycle of test level E shall be modified as follows:

a. The airspeed indicator shall be pneumatically cycled from 0 knots to approximately 3/4 of the airspeed range and return to 0 within approximately 15 minutes for at least 4 cycles per hour.

b. Period B of MIL-STD-781, figure 1 shall be approximately 4 hours.

c. Indicator lighting shall be on 57 minutes and off 3 minutes each hour.

d. During each hour, the indicator shall be subjected to 10 minutes of external vibration of ±0.5g at any frequency between 20 and 60 Hz.

4.6.22.2 Reliability sampling phase tests. Samples of the airspeed indicator shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling-phase, test level E (as modified herein) shall be used. An alternative Production Reliability test may be specified by the procuring activity to be performed for a fixed time period of 250 hours. When the alternative Production Reliability Test Plan is specified in the contract or purchase order, the contractor shall prepare such a plan for procuring activity approval (see 6.2.2). The plan shall consider the specific airspeed indicator design and be based upon the reliability test plans of more recent production contracts. Guidelines will be specified by the procuring activity. Such a plan must be provided for the hold of a production lot until the test is completed.

4.6.22.3 Longevity tests. Unless otherwise specified by the contract or purchase order, a longevity test shall be performed in accordance with MIL-STD-781, test plan XXVIII and the duty cycle herein, except that 3,000 hours operating time shall be accumulated on each of two units. Two of the airspeed indicators used during the reliability assurance tests shall be used for this test and all operating time accrued during reliability testing may be counted toward the longevity time. The airspeed indicator performance checks may be performed at 500 hour intervals in lieu of 250 hour intervals after completion of 1000 hours of testing for each indicator. Data shall be retained for procuring activity approval (see 6.2.2).

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4.6.22.4 Performance characteristics to be measured during longevity and reliability assurance tests. The airspeed indicator shall be observed each working day for any obvious failures such as rough pointer, excessive scale errors, or failure of the integral lighting. At the end of each 250 operating hours, the indicators shall be subjected to the following tests:

a. Case leakage (see 4.6.3).

b. Differential pressure capsule leakage (see 4.6.4).

c. Scale error at room temperature (see 4.6.5).

d. Friction (see 4.6.6.).

e. Integral lighting (see 4.6.8).

5. PACKAGING.

5.1 Packaging and preservation. The airspeed indicator shall be packaged in accordance with MIL-STD-794, level A or C, as specified in contract or order (see 6.2.1). For level A packaging, the method of preservation shall be in accordance with MIL-P-116, method 1A-5 without preservation compound, using reusable metal containers.

5.2 Packing. The indicator shall be packed in accordance with MIL-STD-794, level A, B, or C as specified in contract or order (see 6.2.1).

5.2.1 Level A. The packaged indicator shall be packed in a protective shipping carton conforming to PPP-B-636, class weather resistant, style optional, special requirements. Protective packing material such as polystyrene or other suitable material shall surround the indicator to protect the entire unit including dial glass bezel and a protruding barometric setting knob from damage.

5.2.2 Level B. The package indicator shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The level C packing for the indicator shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for one destination equal 40 cubic feet or more. Unitized load shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Indicators, packed as specified in 5.2.1, shall be unitized on pallets in comformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Indicators, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Indicators, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

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5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification marking specified in the contract (see 6.2.1), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.3.

6. NOTES.

6.1 Intended use. The airspeed indicators are intended for standby or emergency use in high performance aircraft to indicate airspeed. Type III indicators are intended for use primarily in V/STOL type aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Airspeed indicator type (see 1.2)

c. If other than standard parts are to be used (see 3.2.1.1)

d. Whether Reliability Qualification Phase tests are required (see 4.6.22.1).

e. Reliability Sampling Phase tests (see 4.6.22.2) (specify MIL-STD-781 or alternative test plan)

f. Longevity Test (see 4.6.22.3)

g. Maintainability demonstration (see 3.3.6 and 4.6.21)

h. Levels of preservation-packaging and packing required (see 5.1 and 5.2)

i. Warranty decal information, if applicable (see 3.3.9.1)

j. Levels of marking (see 5.3.1)

6.2.2 Data requirements. When this specification is used in an aquisition which incorporates a DD Form 1423, Contract Data Requirement Lists (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL

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incorporated into the contract. When the provisions of DAR-7-104.9(n)(2) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.2.3, 3.2.2.6	Deviations, Request for	UDI-E-23102A
4.5.1.2.1.1 e	Procedure, Manufacture Production Test	DI-T-4903
4.5.1.2.1.1 e	Report, Manufacture Production Test	DI-T-4903
4.5.2.1.1	Report, Reliability (monthly)	DI-R-7035
4.6.22.2,	Procedure, (Alternate) Reliability Assurance Test	DI-R-7035
4.6.22.3	Report, Longevity Test	DI-T-4904
4.5.2.1.1, 4.5.2.1.5	Report, Failure	UDI-R-21141A
4.6.21	Procedure, Maintainability	DI-T-4903

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific aquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Good as new. "Good as new" shall mean the airspeed indicator has been operated less than 10 percent of specified MTBF operation. Prior to delivery, the indicator shall be subjected to and meet all the requirements of a new indicator, except the run-in test need not be performed.

6.3.2 Refurbished. "Refurbished" shall mean the airspeed indicator has been completely overhauled with all component parts meeting current parts standards; and prior to delivery the indicator shall have been subjected to and met all requirements of a new indicator.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Product Lists whether or not such products have actually been so listed by the date. Suppliers are directed to Provisions Governing Qualification SD-6 and are urged to arrange to have the products that they propose to offer to the Federal Government

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tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air System Command, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity.

Preparing Activity:
Navy-AS
Project No. 6610-N053

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-85506(AS)	Indicator, Indicated Airspeed, Standby Types I, II, and III

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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